

SQL Joins and Their Uses

1. INNER JOIN

Use: Returns only the matching rows from both tables.

Syntax:

```
SELECT * FROM table1 INNER JOIN table2 ON table1.id = table2.id;
```

When to use:

Find customers who placed orders.

Match employees with departments.

Retrieve related records from two tables.

2. LEFT JOIN

Use: Returns all rows from the left table and matching rows from the right table. Non-matching rows show NULL.

Syntax:

```
SELECT * FROM table1 LEFT JOIN table2 ON table1.id = table2.id;
```

When to use:

Find customers with or without orders.

Keep all records from the main table.

Identify missing matches.

3. RIGHT JOIN

Use: Returns all rows from the right table and matching rows from the left table.

Syntax:

```
SELECT * FROM table1 RIGHT JOIN table2 ON table1.id = table2.id;
```

When to use:

Keep all records from the right table.

Find unmatched rows in the left table.

4. FULL OUTER JOIN

Use: Returns all rows from both tables. Unmatched rows contain NULL.

Syntax:

```
SELECT * FROM table1 FULL OUTER JOIN table2 ON table1.id = table2.id;
```

When to use:

Compare two tables.

Find matching and non-matching records.

Data reconciliation.

Note: MySQL uses UNION workaround.

5. CROSS JOIN

Use: Returns every possible combination of rows from both tables (Cartesian Product).

Syntax:

```
SELECT * FROM table1 CROSS JOIN table2;
```

When to use:

Generate all possible combinations.

Create product variants.

Test datasets.

6. SELF JOIN

Use: Joins a table with itself.

Syntax:

```
SELECT a.name,b.name FROM employees a JOIN employees b ON  
a.manager_id=b.employee_id;
```

When to use:

Employee–Manager relationships.

Parent–Child hierarchy.

Recursive relationships.

Quick Revision Table

Join	Returns	Common Use
INNER	Only matching rows	Customer orders
LEFT	All left + matching right	Customers without orders
RIGHT	All right + matching left	Keep right records
FULL	All rows	Compare datasets
CROSS	All combinations	Product combinations
SELF	Same table	Employee-Manager

Easy Memory Trick

INNER → Match only ■

LEFT → Keep left table ■

RIGHT → Keep right table ■

FULL → Keep everything ■

CROSS → Every combination ■

SELF → Same table joins itself ■